

A PROJECT TITLE

ON

License Plate Recognition for Traffic Monitoring Using CNN and OCR

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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DEPARTMENT OF CSE

Name of Title: License Plate Recognition for Traffic Monitoring Using CNN and OCR

1. ABSTRACT:

The integration of deep learning with Optical Character Recognition (OCR) is transforming traffic monitoring systems by enabling automated license plate recognition (LPR) in real-world scenarios. CNNs excel at extracting spatial features from license plate images, while OCR ensures precise text recognition, making the combination highly effective for surveillance and smart city applications. To rigorously evaluate this approach, the OpenALPR benchmark dataset—containing diverse license plate images under varying lighting and environmental conditions—will be employed. Performance will be measured using accuracy and F1-score metrics, ensuring both precision and recall are considered. By addressing challenges such as blurred images, angled plates, and poor lighting, the project aims to demonstrate a scalable solution that minimizes manual intervention, enhances monitoring efficiency, and supports the vision of intelligent urban infrastructure.

2. INTRODUCTION

Traffic congestion, road safety, and law enforcement are pressing challenges in today's rapidly urbanizing world. With the number of vehicles increasing every year, manual monitoring methods are no longer sufficient to ensure smooth traffic flow and compliance with regulations. This motivated us to work on a project that uses automation and artificial intelligence to assist in traffic management. By reducing human effort and minimizing errors, such systems can make cities safer, more efficient, and better prepared for future growth.

License Plate Recognition (LPR) is one of the most effective tools in this domain. It enables automatic identification of vehicles by detecting and reading their license plates from images or video feeds. In this project, we combine Convolutional Neural Networks (CNNs) for plate detection with Optical Character Recognition (OCR) for character extraction. CNNs provide robust feature learning to locate plates under varying conditions, while OCR ensures accurate conversion of plate text into digital format. Together, these techniques form a reliable system for

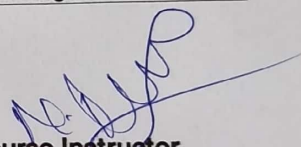
traffic monitoring, violation detection, and vehicle tracking, contributing to the broader vision of intelligent transportation systems.

3. SOFTWARE REQUIREMENTS:

Developing a License Plate Recognition system requires both efficient software tools and suitable hardware resources to ensure realtime performance. The software stack provides the environment for image processing, deep learning model training, and OCR implementation, while the hardware ensures smooth execution of computationally intensive tasks. Together, they form the backbone of the project, enabling accurate detection and recognition of license plates under varying traffic conditions.

Requirement Type	Specification
Software	Python 3.8 or above
	TensorFlow (for CNN implementation)
	OpenCV (for image preprocessing and plate detection)
	NumPy, Pandas (for data handling)
	Matplotlib/ Seaborn (for visualization)
	Tesseract OCR (for character recognition)
Hardware	Standard PC or laptop
	Minimum 8 GB RAM (recommended)
	GPU support optional but useful for faster training


Signature of the Guide


Course Instructor